



Cite this: *New J. Chem.*, 2019, **43**, 16374

Theoretical insights into the competitive metal bioaffinity of lactoferrin as a metal ion carrier: a DFT study†

Bahareh Honarparvar,* Suvardhan Kanchi and Krishna Bisetty *

Metal–protein complexes, specifically lactoferrin (Lf), an iron-binding glycoprotein found naturally in milk and several other body fluids play a pivotal role in all living organisms. Due to the involvement of these metalloproteins in health-related studies, a deeper understanding of the metal ion bio-affinities in the metal binding pocket would be of significant research interest. Accordingly, the broader goals of this study are to assess the competitiveness of the biological metal binding bio-affinities found in microfluidics. This study is therefore aimed at an extensive exploration of the recognition of individual biological metal ions and their interactions with the metal ligating residues of Lf (TYR92, TYR192, HIS253, ASP60, and CO_3^{2-}) in a vacuum and in implicit and explicit aqueous solution at physiological pH. Density functional theory (DFT) with the B3LYP functional was used to identify effective electronic factors that potentially contribute to the extent of the metal ion bioaffinities. The LANL2DZ basis set was used for Ti^{4+} , Fe^{3+} , Cr^{3+} , Co^{2+} , and Cu^{2+} and the 6-31+G* basis set was applied for the rest of the atoms. The metal ion binding energies in a vacuum and in implicit and explicit aqueous media together with $\Delta E_{\text{HOMO-LUMO}}$ chemical hardness (η) and softness (S) revealed the strongest metal affinity for Ti^{4+} in comparison with others. The observed trends for metal affinity follow the order $\text{Ti}^{4+} > \text{Fe}^{3+} > \text{Cr}^{3+} > \text{Co}^{2+} > \text{Cu}^{2+}$, which is supported by the reported experimental binding constant values ($\log k_{\text{binding}}$) along with ICP-mass spectroscopy in the literature. The stabilization energies derived using the second order perturbation theory, $E^{(2)}$, of NBO analysis confirmed the greater charge transfer for the observed trends in the metal binding. Furthermore, the observed metal binding energies in the explicit solvation model ($\Delta E_{\text{explicit}}$) with six water molecules coordinated to the metal ions in the octahedral system highlight the importance of the hydration of free metal ions in the metal binding process. The findings of this study identify efficient electronic factors as major contributors to the metal binding affinities, with promising possibilities for the design of metalloproteins and sensing of the metal ions.

Received 21st July 2019,
Accepted 21st September 2019

DOI: 10.1039/c9nj03786a

rsc.li/njc

Introduction

Metal ions particularly Ti^{4+} , Fe^{3+} , Cr^{3+} , Co^{2+} , and Cu^{2+} in human serum are vital elements for metal binding and biological ion transportation processes. They are also key elements for many pathogenic microorganisms.^{1–3} The mechanism of the metal ion uptake relies on the bio-availability of the metal ions or in the presence of metalloproteins which bind to metal ions.⁴ A deeper understanding of the role of metal ions on the structure and function of lactoferrin is therefore crucial for health, and disease and therapy diagnosis.⁵ Iron uptake by

transferrin family proteins will make this element inaccessible for pathogenic microorganisms, preventing them from further development. Since the formation of free radicals which can be generated due to the catalytic role of metals in an unbounded state generates many other harmful biochemical processes,^{6,7} iron acquisition by lactoferrin can potentially protect the living organisms from reactive oxidative stress.^{6,8,9}

Human lactoferrin (Lf) is an 80 kDa molecular weight iron-binding glycoprotein with two homologous metal binding sites (N and C lobes), initially found in milk and secretion tears, binds ferric and other metal ions with high affinity.^{10–16} The potent iron-binding characteristics of lactoferrin lead to iron deficiency, which is a key factor in host defence. Due to its antibacterial, antiviral, antifungal, and anticancer properties, lactoferrin is also effective for wound healing and bone growth, or it can act as an iron carrier.^{12,17,18} It is notable that the majority of the metalloproteins have transition metal cofactors¹⁹

Department of Chemistry, Durban University of Technology, P.O. Box 1334, Durban 4000, South Africa. E-mail: bisettyk@dut.ac.za, baha.honarparvar@gmail.com

† Electronic supplementary information (ESI) available. See DOI: 10.1039/c9nj03786a

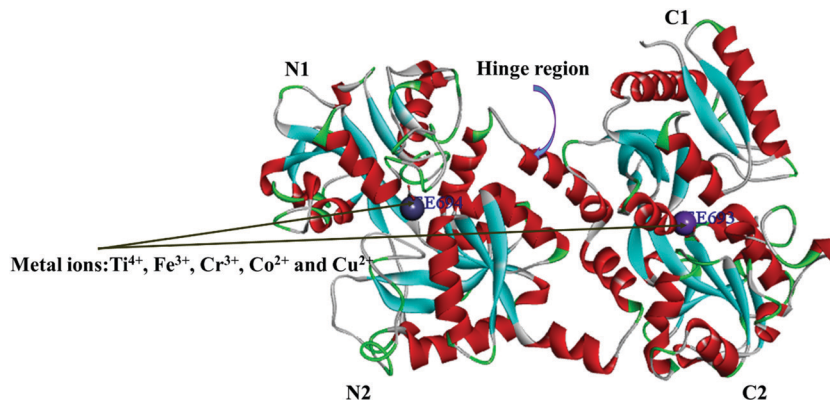


Fig. 1 Graphical representation of the different human lactoferrins generated using the Discovery studio. The position of iron ions in each lobe is marked in CPK in purple.

as crucial nutrients for all organisms to facilitate the chemical transformations in metabolism and detoxification, and charge transfer centers.²⁰

A key structural feature of Lf is the 2-fold internal primary structural repeat units, which is also apparent in proteins connected by the hinge region and the iron binding sites.⁵ The N-terminal lobe of lactoferrin (residues 1–333) is homologous to its C-lobe (residues 345–691) (Fig. 1) with a high degree of sequence identity, and while this lobe regulates the enhanced antimicrobial effect, the C-lobe contributes to various therapeutic functions.^{21–24}

Due to the sequence similarity of the two lobes of Lf in this study, the metal binding to the N-lobe was undertaken to compare the binding affinities of the (Ti^{4+} , Fe^{3+} , Cr^{3+} , Co^{2+} , and Cu^{2+}) metal ions to Lf.

Advances in analytical techniques mainly electron paramagnetic resonance (EPR) spectroscopy, inductively coupled plasma-mass spectrometry (ICP-MS) and field-flow fractionation (FFF) are state-of-the-art techniques that have greatly contributed to characterization studies involving metalloproteins and protein-metal complexes. EPR can potentially be used to study the electronic absorption spectra of metalloproteins, yielding molar absorptivity per bound metal ion ($\Delta\epsilon_i$) and stability constants ($\log K_{\text{binding}}$) for metal binding assessments.⁵ EPR spectroscopy was used to study Fe^{3+} and Cu^{2+} of transferrin complexes,⁵ which have the same binding residues around the metal ions.^{24,25}

On the other hand, ICP-MS is another powerful and sensitive analytical method mainly used for elemental quantification and metal-binding protein determination in metal-protein complexes.^{26,27}

Moreover, ICP-MS was used to identify the bioavailability of wear ^{65}Cu , ^{53}Cr , ^{59}Co and ^{98}Mo metals and metalloproteins in biofluids from hip replacement patients to assess the metal protein binding. The observed trends for the metal bioaffinities follows the order $\text{Cr}^{3+} > \text{Co}^{2+} > \text{Cu}^{2+}$.²⁸

However, the coupling of FFF with ICP-MS is a powerful technology for the separation and characterization of protein-metal complexes. Due to the high biocompatibility caused by its resistance to corrosion by body fluids such as serum and urine, amongst different metal ions, titanium (Ti) is an

abundant element in implants, prosthetics, and artificial hip joints^{2,3} with various medicinal applications including known anticancer drugs. Based on the very limited data on the Ti metabolism, some proteins like Lf might be involved in the binding mechanism of Ti,²⁹ which demands a detailed electronic structure investigation.

The effective physico-chemical features related to the functions of bovine lactoferrin were studied by Rosa *et al.*,¹² and iron-saturation was found to be effective on thermal stability and resistance to proteolysis. They also reported that $\text{Al}(\text{III})$, $\text{Cu}(\text{II})$, $\text{Mg}(\text{II})$, $\text{Mn}(\text{II})$, and $\text{Zn}(\text{II})$ metal ions are chelated by Lf, with a lower affinity than $\text{Fe}(\text{III})$.¹² In terms of the binding mechanism, it is described that iron is weakly bound and consequently destabilizes the hinge segment (Fig. 1) with significant conformational fluctuations resulting in the formation of a stable iron binding site with a stable closed conformation.³⁰ The biological activities of different bovine Lf forms such as apoLf, holoLf, and MnLf that differ in their metal saturation and depletion levels were also assessed and compared with the naturally occurring (native) Lf.¹³

Acosta *et al.*, applied size exclusion chromatography coupled to ICP-MS for metalloprotein determination in human breast milk involving Mn, Co, Cu and Se quantification. They performed accurate analysis on metalloproteins with satisfactory sensitivity using size exclusion chromatography (SEC) coupled to ICP-MS. They proposed a valid strategy for the determination of metals and metalloproteins. Notably the presence of Mn, Co, Cu and Se bound to two protein fractions in the whey of human breast milk (HBM), and the observed metal levels were consistent with those reported in the literature.³¹ Furthermore, the binding of human apo-lactoferrin with the $\text{Ti}(\text{IV})$ complex was performed using spectrophotometric and spectrofluorometric techniques.¹⁶ Rapid and tight binding of trivalent chromium (Cr^{3+}) to serum transferrin in the presence of 25 mM bicarbonate (CO_3^{2-}) was observed using *in vivo* studies. However, the binding mechanism at a molecular level is unknown,¹¹ which is the main thrust of the current work.

The low concentrations coupled with the possible spectral interferences made accurate identification of the exact bioaffinities and increased particularly Ti in human serum by experimental

measurements are quite challenging.² Moreover, the capability of lactoferrin in terms of its potential binding to a range of metal ions^{10–13,15,16,32,33} has been the driving force for the implications of Lf in the diagnosis and treatment of various diseases. Several experimental studies on the structure and function of lactoferrin^{7,12,13,16–18,21,23,31,34–41} as well as some computational studies have been performed to address the structure and function of Lf.^{22,39,42–46}

From a computational viewpoint, the molecular structure, binding and dynamics of human Lf, and of the Lfs from cows, horses, buffalos and camels, were discussed by Baker and coworkers. A highly conserved three-dimensional structure with differences between regions was observed. Molecular dynamics (MD) studies revealed that the iron uptake and release occurred by domain movements which led to conformational changes affected by cooperative interactions between the lobes.²²

Despite the aforementioned studies, to the best of our knowledge, no systematic calculations have been performed to identify the competitive bioaffinities of the aforementioned metal ions in the metal sphere of Lf.

According to the literature and crystallographic studies, different metal ions are ligated to the same metal binding residues together with a carbonate ion in a six-fold (octahedral) coordination pattern.^{11,30,47–49} In the case of Cu^{2+} ions, amongst the C and N-lobes of lactoferrin, they appear in distorted octahedral coordination caused by the Jahn–Teller distortion (four short bonds and two longer apical bonds).^{4,50–52} It was concluded that the bicarbonate (CO_3^{2-}) anion induced a change in the coordination geometry of the copper ions at the N-site.⁵¹ There is a possibility that the extra space in the C-lobe site of Lf allows the Cu^{2+} ions to undergo a five-coordinate square pyramidal geometry within small movements of the metal, carbonate and other ligands.⁵³ To make a consistent comparison to probe the metal binding affinity, the N-lobes of lactoferrin with six-fold (octahedral) coordination patterns were used in this study. The present study explores the electronic structure analysis of the selected metal ions substituted in the binding pockets of human serum lactoferrin at the molecular level and the binding affinities between the metal ions and the BS (binding site) groups of Lf in a vacuum, and in implicit and explicit aqueous media. Also, we calculate the bond lengths between the ligating atoms to the metal ions and the frontier molecular orbitals ($\Delta E_{\text{HOMO-LUMO}}$), chemical hardness (η) and softness (S). Furthermore, the potency of charge transfer between the selected first row transition metal ions with the ligating residues (TYR92, TYR192, HIS253, ASP60, and CO_3^{2-}) is probed based on the stabilization energies, using the second order $E^{(2)}$ perturbation theory of natural bond orbital (NBO) analysis.

Computational details

System preparation

The X-ray crystallographic structure of human lactoferrin (PDB code: 1B0L)⁴¹ consisting of a polypeptide chain of 691 residues was derived from the Protein Data Bank (PDB).⁵⁴ The protonation state of the ionizable groups of the protein residues was assigned

pK_a calculations under physiological conditions (pH 7.4) using the H++ web server.⁵⁵ It is notable that, based on the protocol for pK_a calculations using the H++ server,⁵⁶ all water molecules and metal ions as well as CO_3^{2-} were removed as non-standard residues and the protonation states were assigned for all titratable Lf residues. The coordinates of the metal ions as well as CO_3^{2-} were added to the generated pdb along with the correct protonation state. The pK_a values computed using the H++ server for the main residues are provided in the ESI.†

Geometry optimization and bond length measurement

The metal binding site (BS) of the N-lobe lactoferrin including the BS residues (carbonate ion (CO_3^{2-}), His253, Asp60, TYR92 and TYR192, Fig. 2) covalently coordinated to the selected Ti^{4+} , Fe^{3+} , Cr^{3+} , Co^{2+} , and Cu^{2+} metal ions was used for metal binding energy calculations. The Lf model was geometry optimized with the DFT method⁵⁷ using the Gaussian 16 software package.⁵⁸ In an earlier DFT study on transferrin (Tf) with the same ligating residues as lactoferrin to the different metal ions in the octahedral system, Sakajiri *et al.*⁵² compared the experimental metal binding constants with DFT-based binding energies. Their findings revealed that the complex structure (whole Tf with metal ions) was not perturbed in the presence of the rest of the protein. They concluded that the metal binding site (BS) model was adequate for the theoretical calculation for probing the metal binding affinities. Therefore, in our study, the metal binding site (BS) of the N-lobe lactoferrin covalently coordinated to the selected Ti^{4+} , Fe^{3+} , Cr^{3+} , Co^{2+} , and Cu^{2+} metal ions was used for DFT studies to assess the metal binding energy affinities.

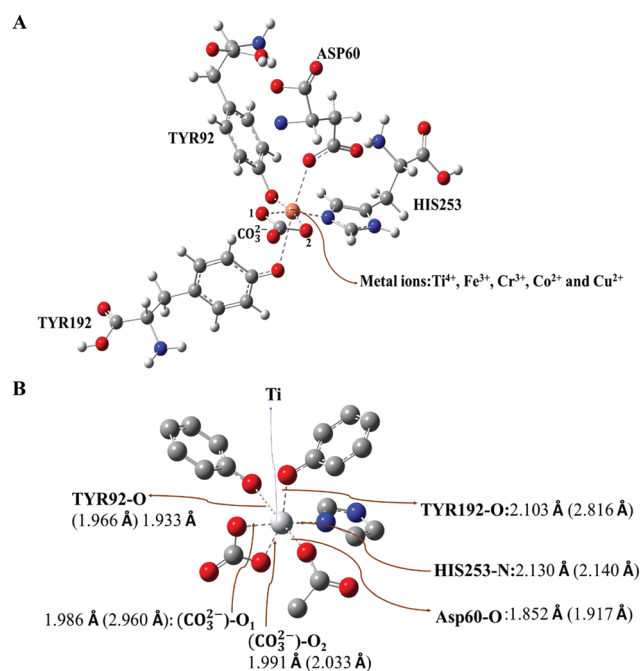


Fig. 2 The optimized structure of (A) the N-lobe of human lactoferrin, from the crystal structure (PDB code: 1B0L),⁴¹ (B) metal ligated atoms and bond lengths for Ti^{4+} –Lf in a vacuum and implicit water (in parenthesis).

B3LYP density functional and LANL2DZ (Los Alamos National Laboratory 2 Double-Zeta)⁵⁹ were applied for the selected transition metal ions, a commonly used ECP type basis set to model the metal atoms⁶⁰ and the 6-31+g* basis set for the rest of the atoms. It is notable that all the selected ions are ligated to the aforementioned residues of Lf in an octahedral configuration with no constraints set during the geometry optimization. Frequency calculations were performed on the optimized systems at the same level of theory using normal mode analysis⁶¹ to ascertain that all structures were at local minima without negative frequencies. The total binding entropies and the individual contributions (translational, rotational, and vibrational) in a vacuum ($\Delta S_{\text{binding}}^{\text{vacuum}}$) were derived from frequency calculations.^{62–64}

The metal–ligand bond lengths involved in the binding pockets were measured in a vacuum and implicit water to assess the contributing factor for the metal-binding bioaffinities.

Binding energy calculations

To gain insight into the metal binding strength of lactoferrin, the metal binding energies ($\Delta E_{\text{binding}}^{\text{vacuum/water}}$) were calculated employing the BS model shown in Fig. 2 in a vacuum and in implicit water as:^{62,64,65}

$$\begin{aligned}\Delta E_{\text{binding}}^{\text{vacuum/water}} &= E_{\text{ion-LfBS}}(\text{vacuum/implicit water}) \\ &\quad - E_{\text{BS of Lf}}(\text{vacuum/implicit water}) \\ &\quad - E_{\text{ion}}(\text{vacuum/implicit water})\end{aligned}\quad (1)$$

The gas phase binding energies also known as the intrinsic metal-binding energies (ΔE_i) were determined under no hydration conditions for the free metal ions with a dielectric constant of $\epsilon = 1$ (in a vacuum).

To assess the bioaffinities of the aforementioned metal ions in bulk water, the metal-binding energies in implicit water ($\Delta E_{\text{implicit}}$) were also calculated in the implicit water Polarizable Continuum Model (PCM),⁶⁶ using the integral equation-formalism polarizable continuum model (IEF-PCM)^{67,68} with a dielectric constant of $\epsilon = 78.39$.

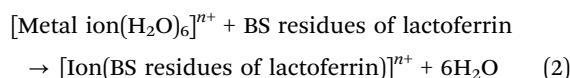
Explicit solvation model

To assess the extent of interference of explicit water molecules in the interactions between the metal ions and the BS residues in the Lf octahedral coordinated system, the metal binding energies in the explicit solvation model ($\Delta E_{\text{explicit}}$) were calculated. In this way, the contribution of the hydration of the free metal ions in aqueous solution to the metal binding was also probed. As in the BS of lactoferrin, the selected transition metals are covalently bound to six ligands (bidentate carbonate ion (CO_3^{2-}), HIS253, Asp60, TYR92 and TYR192); these 6 ligating groups were removed and replaced with 6 water molecules and the solvation energies of each free metal ion and the metal binding energy were calculated in the explicit hydration model ($\Delta E_{\text{explicit}}$). Adeowo *et al.*,⁶³ studied the significance of ion–water interactions in the chelation interaction energies of Cu^{2+} , Ga^{3+} , Sc^{3+} , and In^{3+} radiometal ions with 1,4,7-triazacyclononane-1,4,7-triacetic acid (NOTA) in solution. In their chelation reaction, six

water molecules were coordinated to the radiometal ions in an octahedral conformation. Bowron *et al.*,⁶⁹ identified the preferred local geometry of the Cu^{2+} aqua-ion hydration shell for the solutions of $\text{Cu}(\text{ClO}_4)_2$ as mixed octahedral, tetrahedral, and trigonal bipyramidal coordination patterns using H/D isotopic substitution neutron scattering and X-ray scattering. As in the BS of lactoferrin, the selected transition metals are covalently bound to the six ligands (bidentate carbonate ion (CO_3^{2-}), HIS253, Asp60, TYR92 and TYR192). Thereafter, the six ligating groups were removed and replaced with six water molecules and the solvation energies of each free metal ion and the metal binding energies were calculated in the explicit hydration model ($\Delta E_{\text{explicit}}$). Furthermore, the hydration energies of free Fe(III), Ga(III), Al(III), Cu(II), Zn(II), and Ni(II) metal ions were calculated with a metal ion coordinated by six water molecules in the octahedral conformation as a solvation model using the DFT method.⁵²

The same six-fold (octahedral) protocol for the solvation model previously applied for different metal ions (including Cu^{2+}) was implemented for the calculation of the metal binding energy in the explicit hydration model ($\Delta E_{\text{explicit}}$).

The metal-binding complexation in solution is defined according to the following reaction:⁶³



where n is the charge of the metal ions ($n = 4$ for Ti(IV); 3 for Fe(III), Cr(III) and 2 for Co(II), and Cu(II)) and six is indeed the coordination number of water molecules to the metal ions in the octahedral system.

The interaction energy between the metal ions and the six water molecules ($E_{\text{int}}^{6\text{H}_2\text{O-ion}}$) is defined according to the equation:⁶³

$$E_{\text{int}}^{6\text{H}_2\text{O-ion}} = E_{6\text{H}_2\text{O-ion}} - E_{\text{ion}} + E_{6\text{H}_2\text{O}}\quad (3)$$

Due to the competitive interactions between BS residues—ion and water—ion, the metal-binding energy in explicit water ($\Delta E_{\text{explicit}}$) includes the following components and is defined as:

$$\Delta E_{\text{explicit}} = E_{\text{binding}}^{\text{vacuum}} - E_{\text{int}}^{6\text{H}_2\text{O-ion}}\quad (4)$$

Natural bond orbital (NBO) calculations

The natural bond orbital (NBO) analysis^{70–72} was carried out using the NBO 3.1 program implemented in the Gaussian16 to calculate the energy eigenvalues of the frontier molecular orbital ($\Delta E_{\text{LUMO-HOMO}}$) energies for the BS of lactoferrin with the selected transition metal ions to assess the stability of the metal binding pocket of Lf which is an indicator of metal ion bioaffinities.

Based on the molecular orbital theory, the Koopmans theorem⁷³ and the finite difference approximation, Parr and Pearson⁷⁴ has defined the chemical hardness (η) as:^{75–78}

$$\eta = \frac{1}{2}(\text{IE} - \text{EA})\quad (5)$$

where IE and EA are the vertical ionization potential and vertical electron affinity, respectively. According to the Koopmans

theorem.⁷³ According to the Kohn–Sham formalism of DFT,⁷⁹ the energy of the highest occupied orbital (E_{HOMO}) is referred to as the ionization potential (IE) and the lowest unoccupied orbital (E_{LUMO}) to electron affinity:⁸⁰

$$\eta = \frac{1}{2}(E_{\text{HOMO}} - E_{\text{LUMO}}) \quad (6)$$

Chemical softness (S), which is an indicator of the electron localization over the covalent bonding of the BS of the lactoferrin–metal ion complexes, is defined as:^{81,82}

$$S = 1/2\eta \quad (7)$$

Second perturbation theory

The stabilization energy, $E^{(2)}$, deduced from the second order perturbation theory analysis of the Fock Matrix,⁸³ was determined on the fully optimized systems to identify the electron donor and acceptor regions of the complexes and the potency of charge transfer from one molecular orbital to another. This electron donation results in stabilization energy ($E^{(2)}$). The stabilization energy $E^{(2)}$ that arises from $i \rightarrow j$ delocalization is defined as follows:

$$E^{(2)} = \Delta E_{ij} = q_i \frac{F(i,j)^2}{\varepsilon_j - \varepsilon_i} \quad (8)$$

where q_i refers to the i th donor orbital occupancy, ε_i and ε_j are the diagonal elements (orbital energies), and $F(i,j)$ is the off-diagonal element. The higher the value of $E^{(2)}$, the more the inter-molecular orbital interactions and consequently greater charge transfer between the electron donors and the electron acceptors (between the binding site ligating groups of the lactoferrin and metal ions),⁸⁴ which leads to complex stability and stronger metal-binding bioaffinities.

Results and discussion

For the comparison of the metal binding strength of the binding pocket for the lactoferrin–ion complexes, lucid data interpretation of the electronic structure properties including binding energies in a vacuum, and implicit and explicit aqueous media is discussed.

Analysis of metal–ligand bond lengths

The optimized structure of the N-lobe human lactoferrin with its metal binding site with key residues is shown in

Fig. 2. Although no constraints were set during geometry optimization, a 3D visual inspection revealed that the shape of the binding site was conserved in an octahedral configuration. Chemical modification, titration and spectroscopic methods on lactoferrin revealed that each lobe of the Lf metal ions is coordinated into two tyrosines (TYR92 and TYR192), a histidine (HIS253) and an aspartic acid (ASP60) with a bidentate carbonate (CO_3^{2-}) as a synergistic anion.^{24,25}

Analysis of metal–ligand bond lengths is an indicator to rationalize the strength of the metal to ligating residues of lactoferrin. According to X-ray crystallography of transferrin family metalloproteins, the metal–ligand bond lengths are around 1.9–2.2 Å.⁵ It was also reported that the synergistic bidentate anion (CO_3^{2-}) plays an important role in creating the metal binding site, and the absence of this anion leads to weaker metal binding.⁵ Interestingly, it was also realized that the strength of the anion binding to transferrin was rationalized and found to correlate with the charge-to-radius ratio.⁸⁵

The measured bond lengths between each metal ion and its ligating residues, namely, TYR92, TYR192, HIS253, ASP60 and CO_3^{2-} on the optimized metal binding site pocket of the N-lobe of the human lactoferrin in Fig. 2 are listed in Table 1.

As shown in Table 1, the optimized bond lengths of the Lf complexes are in the expected range of the crystal structures⁵ for the transferrin family metalloproteins. The incorporation of the implicit water to the Lf complexes led to an increase in bond lengths between the Ti^{4+} , Fe^{3+} , Cr^{3+} , Co^{2+} , and Cu^{2+} metal ions and the ligating residues involved in the metal binding pockets of (Lf) when compared with those in a vacuum. The values for the bond lengths in aqueous solution are closer to the Fe^{3+} –Lf crystal structure marked in bold. Furthermore, particularly for metal ions with higher oxidation states the charge-to-radius ratios are correlated with their ionic radii with a greater ratio obtained for Ti^{4+} followed by Fe^{3+} and Cr^{3+} .

Binding energies

The binding energy values provide a measure of the bioaffinities of Ti^{4+} , Fe^{3+} , Cr^{3+} , Co^{2+} , and Cu^{2+} metal ions to the BS of Lf. The binding energies of the Lf complexes in a vacuum and in an implicit water solvation model derived from eqn (1) are listed in Table 2. The total entropies and the individual contributions [translational (ΔS_{trans}), rotational (ΔS_{rot}), and vibrational (ΔS_{vib})] in a vacuum ($\Delta S_{\text{binding}}^{\text{vacuum}}$) for the binding of the selected metal ions with the ligated residues of Lf in a vacuum ($\Delta S_{\text{binding}}^{\text{vacuum}}$) were obtained from the frequency calculations to better understand

Table 1 The optimized bond lengths and experimental crystal structure values (PDB code: 1B0L)⁴¹ (marked in bold) between the Ti^{4+} , Fe^{3+} , Cr^{3+} , Co^{2+} , and Cu^{2+} metal ions and the ligating residues involved in the metal binding pocket of (Lf) in a vacuum and implicit water (in parenthesis). The selected bond lengths (Å) are labelled in Fig. 2

Complexes	Spin	HIS253-N	Asp60-O	TYR92-O	TYR192-O	(CO_3^{2-})-O ₁	(CO_3^{2-})-O ₂	Ionic radius ⁵ (Å)	Charge/radius
Ti^{4+} –Lf	1	2.130 (2.140)	1.852 (1.917)	1.933 (1.966)	2.103 (2.816)	1.986 (2.960)	1.991 (2.033)	0.605	6.611
Fe^{3+} –Lf	6	2.154 (2.293)	2.038 (2.929)	1.864 (2.007)	1.862 (2.145)	2.080 (2.093)	2.930 (2.953)	0.650	4.615
Fe^{3+} –Lf crystal		2.120	2.175	2.044	1.869	2.026	2.163		
Cr^{3+} –Lf	4	2.063 (2.077)	1.931 (1.940)	2.000 (2.059)	2.048 (2.971)	1.967 (1.968)	1.993 (2.002)	0.520	5.769
Co^{2+} –Lf	4	1.965 (2.024)	1.942 (2.071)	1.884 (1.931)	2.257 (2.692)	1.903 (2.010)	1.921 (1.981)	0.745	2.684
Cu^{2+} –Lf	2	1.907 (1.956)	2.009 (2.239)	1.961 (1.984)	1.558 (2.418)	1.840 (1.984)	2.006 (2.357)	0.730	2.739

the influence of the possible entropic effects on the metal binding.

The most negative binding energies indicate a stronger metal bioaffinity both in a vacuum and in implicit water following the trend $\text{Ti}^{4+} > \text{Fe}^{3+} > \text{Cr}^{3+} > \text{Co}^{2+} > \text{Cu}^{2+}$. It was revealed that the presence of water leads to stronger metal bioaffinities, which indicates that water plays a catalytic role in stabilizing the binding pocket of Lf. The presence of metal ions in the Lf's binding pocket induces conformational changes during geometry optimization, which is indeed apparent in the reported bond lengths in Table 1 and consequently leads to different metal binding energies. As listed in Table 2, the complexation interaction energies for complexes ($E_{\text{ion-LfSBS}}$) are considerably more negative and consequently more stable than those of the complexes in a vacuum. This observation shows that water could stabilize the complexation within the binding pocket. The total and translational metal-binding entropies (ΔS_{tot} and ΔS_{trans}) follow the order of metal bioaffinities as $\text{Ti}^{4+} > \text{Fe}^{3+} > \text{Cr}^{3+} > \text{Co}^{2+} > \text{Cu}^{2+}$, which matches the trend of the metal binding energy values and highlights the loss of entropies (more negative values) for stronger metal binders. The rotational and vibrational entropies (ΔS_{rot} and ΔS_{vib}) appeared to be more positive down the ion-Lf series. Among the reported entropic contributions, the vibrational entropy (ΔS_{vib}) seems to be the driving parameter that controls the total entropies (ΔS_{tot}) of the lactoferrin-ion complexes. This observation indicates that the presence of the ion metals induces conformational changes as reflected by the reported entropic components of the binding free energies.

To calculate the binding energies of the Ti^{4+} , Fe^{3+} , Cr^{3+} , Co^{2+} , and Cu^{2+} metal ions to lactoferrin in explicit aqueous media, the optimized structures of lactoferrin-ion complexes, the metal ions surrounded by 6 water molecules and the free water clusters are shown in Fig. 3. The binding energies of the

complexes in a vacuum and explicit water based on eqn (2)–(5) are listed in Table 3.

The same in terms of metal ion bioaffinity in explicit water was observed as $\text{Ti}^{4+} > \text{Fe}^{3+} > \text{Cr}^{3+} > \text{Co}^{2+} > \text{Cu}^{2+}$. It seems that the charge or oxidation states of the selected metal ions play an important role in the binding. The obtained trend is consistent with the earlier observation that Al^{3+} , Cu^{2+} , Mg^{2+} , Mn^{2+} , and Zn^{2+} metal ions were chelated by Lf, with a lower affinity than Fe^{3+} .¹² Furthermore, our results are supported by the experimentally measured binding constant values ($\log K_{\text{binding}}$)⁵ for Fe^{3+} ($\log K_{\text{binding}} = 21.4 \text{ M}^{-1}$) $>$ Cu^{2+} ($\log K_{\text{binding}} = 12.3 \text{ M}^{-1}$) and the reported pick population and sharpness in the ICP-MS results for $\text{Cr}^{3+} > \text{Co}^{2+} > \text{Cu}^{2+}$.²⁸ The lower values of the metal-binding energies ($\Delta E_{\text{binding}}$) in implicit aqueous solution (in parentheses) in comparison to vacuum reveal that in the presence of water, the metal binding energies of all Lf-metal ion systems are significantly less negative than those in the absence of water, which implies that the interaction of the Lf-metal ions in bulk water (implicit solution in Table 2) as well as in explicit water molecules in Table 3 is less favourable and less stable. Indeed, the presence of implicit/implicit water might compete with Lf for interaction with the ions, and therefore reduce the metal binding affinities in both implicit and explicit aqueous solutions.

Natural bond orbital (NBO) analysis

The calculated DFT-based quantities such as frontier molecular orbital obtained orbital ($\Delta E_{\text{HOMO-LUMO}}$) energies, chemical hardness (η) and softness (S) for the transition metal ions (Ti^{4+} , Fe^{3+} , Cr^{3+} , Co^{2+} , and Cu^{2+}) of the complexes are listed in Table 4.

The higher ($\Delta E_{\text{HOMO-LUMO}}$) energy gap, higher chemical hardness (η) (hence the highest value of chemical softness^{81,82}) and lower softness (S) listed in Table 4 are attributed to the stability of the metal binding pocket of lactoferrin which affects the extent of

Table 2 The metal-binding energies ($\Delta E_{\text{binding}}$) of the Ti^{4+} , Fe^{3+} , Cr^{3+} , Co^{2+} , and Cu^{2+} metal ions involved in the BS model of lactoferrin (Lf) (Fig. 2) in a vacuum and in an implicit water solvation model (in parentheses) and total entropies (ΔS_{tot}) and the individual contributions [translational (ΔS_{trans}), rotational (ΔS_{rot}), and vibrational (ΔS_{vib})] obtained by B3LYP/6-31+G* and LANL2DZ basis sets for ions based on eqn (1)

Complexes	$E_{\text{ion-LfSBS}}$ (Hartree)	$E_{\text{ion-LfSBS}}$ (Hartree)	E_{ion} (Hartree)	$\Delta E_{\text{binding}}^{\text{vacuum/water}}$ (Hartree)	$\Delta E_{\text{binding}}^{\text{vacuum/water}}$ (kcal mol ⁻¹)	$\Delta S_{\text{binding}}^{\text{vacuum}}$ (cal mol ⁻¹ K ⁻¹)
Lf-Ti ⁴⁺	-2720.3710 (-2789.0118)	-1700.5774 (-1799.0371)	-232.2580 (-234.7553)	-787.5354 (-755.2193)	-494 178 (-473 900)	$\Delta S_{\text{tot}} = -33.434$ $\Delta S_{\text{trans}} = -16.885$ $\Delta S_{\text{rot}} = 0.352$ $\Delta S_{\text{vib}} = -16.901$
Lf-Fe ³⁺	-2704.3506 (-2754.3200)		-385.0498 (-387.1279)	-618.7233 (-568.1549)	-388 248 (-356 517)	$\Delta S_{\text{tot}} = -31.630$ $\Delta S_{\text{trans}} = -17.947$ $\Delta S_{\text{rot}} = 0.428$ $\Delta S_{\text{vib}} = -13.228$
Lf-Cr ³⁺	-2607.2204 (-2667.2538)		-306.1363 (-308.1648)	-600.5066 (-560.0518)	-376 817 (-351 433)	$\Delta S_{\text{tot}} = -29.543$ $\Delta S_{\text{trans}} = -18.384$ $\Delta S_{\text{rot}} = 0.601$ $\Delta S_{\text{vib}} = -11.760$
Lf-Co ²⁺	-2698.9959 (-2699.1084)		-430.7792 (-431.9256)	-567.6392 (-468.1456)	-356 193 (-293 761)	$\Delta S_{\text{tot}} = -24.859$ $\Delta S_{\text{trans}} = -20.294$ $\Delta S_{\text{rot}} = 0.780$ $\Delta S_{\text{vib}} = -5.345$
Lf-Cu ²⁺	-2756.8993 (-2729.1357)		-524.2965 (-524.9175)	-532.0291 (-405.1810)	-333 848 (-254 251)	$\Delta S_{\text{tot}} = -22.663$ $\Delta S_{\text{trans}} = -21.986$ $\Delta S_{\text{rot}} = 0.937$ $\Delta S_{\text{vib}} = -1.614$

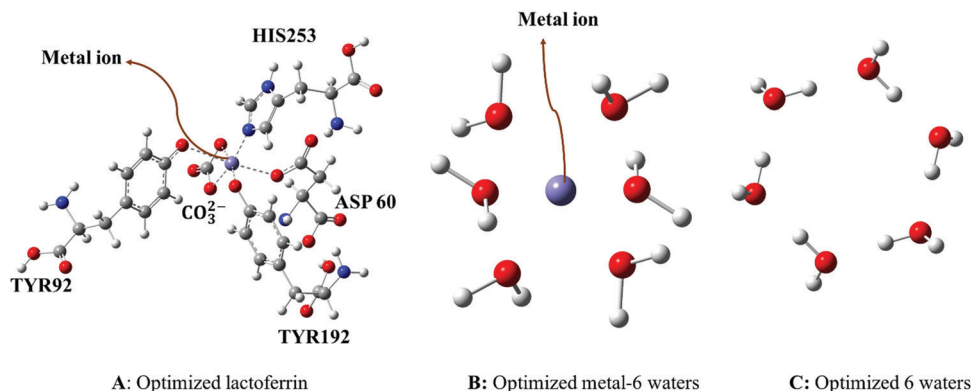


Fig. 3 The optimized structures of lactoferrin–ion complexes, metal ions surrounded by six water molecules and free water clusters.

Table 3 The binding energies between Ti⁴⁺, Fe³⁺, Cr³⁺, Co²⁺, and Cu²⁺ metal ions and BS residues of lactoferrin (Lf) (Fig. 3) in explicit aqueous solution ($\Delta E_{\text{explicit}}$) and their corresponding components defined in eqn (2)–(4) and obtained by B3LYP/6-31+G* and LANL2DZ basis sets for ions

Complexes	$\Delta E_{\text{binding}}^{\text{vacuum}}$ (Hartree)	$E_{\text{int}}^{\text{6H}_2\text{O-ion}}$ (Hartree)	$\Delta E_{\text{explicit}}$ (kcal mol ^{−1})
Lf-Ti ⁴⁺	−787.5354	−781.2458	−3946
Lf-Fe ³⁺	−618.7233	−613.4886	−3284
Lf-Cr ³⁺	−600.5066	−596.0052	−2824
Lf-Co ²⁺	−567.6392	−596.0052	−2294
Lf-Cu ²⁺	−532.0291	−528.4772	−2228

metal ion bioaffinities. This observation is in line with the established fact that the chemical hardness decrease will lead to an increase in the chemical stability of the system,⁸⁶ which in our case, the system is indeed the binding pocket of Lf. The obtained result is also in agreement with the trend of $\Delta E_{\text{HOMO-LUMO}}$ gaps which is also an indicator of chemical stability, *i.e.*, the greater $\Delta E_{\text{HOMO-LUMO}}$ gap provides less chance for electron excitation and makes the system less reactive and more chemically stable.^{75,80} Undoubtedly, the stability of the metal binding pocket of lactoferrin depicted in Table 4 will electronically contribute to the observed metal ion bioaffinities in Lf and reaffirm the reported metal binding energies in a vacuum, and in implicit and explicit aqueous solutions (Tables 2 and 3).

Analysis of charge transfer using the second perturbation theory

The molecular orbital charge transfer between the Ti⁴⁺, Fe³⁺, Cr³⁺, Co²⁺, and Cu²⁺ metal ions and the ligating residues

Table 4 The frontier molecular obtained orbital ($\Delta E_{\text{HOMO-LUMO}}$) energies, chemical hardness (η) and softness (S) for the transition metal ions (Ti⁴⁺, Fe³⁺, Cr³⁺, Co²⁺, and Cu²⁺) involved in the binding pocket of lactoferrin (Lf) derived by eqn (6) and (7) obtained by B3LYP/6-31+G* and LANL2DZ basis sets for ions

Complexes	$\Delta E_{\text{HOMO-LUMO}}$ (eV)	η (eV)	S (eV)
Lf-Ti ⁴⁺	9.524	4.762	0.104
Lf-Fe ³⁺	4.957	2.478	0.201
Lf-Cr ³⁺	3.863	1.931	0.258
Lf-Co ²⁺	2.010	1.005	0.497
Lf-Cu ²⁺	1.094	0.547	0.913

involved in the metal binding pocket of (Lf) derived by the second perturbation theory of NBO analysis is of importance because it influences the bioaffinities of the metal ions. The second-order Fock matrix involved in the second perturbation theory is helpful for evaluating the donor–acceptor interactions in the system.^{87,88} The larger the stabilization energy ($E^{(2)}$), the higher the possibilities of the charge transfer between the ligating residues and the metal ions which leads to stronger metal ion binding.^{89–91}

As shown in Table 5, different possibilities of inter-molecular orbital donor → acceptor charge transfer can be noticed between the $\sigma_{\text{C-O}}$ (ASP60), $\sigma_{\text{C-O}}$ (TYR92), $\sigma_{\text{C-O}}$ (TYR192), $\sigma_{\text{C-N}}$ (HIS253) and $\sigma_{\text{C-O1,2}}$ (CO₃^{2−}) and their corresponding lone pairs (LP) to LP* of the Ti⁴⁺, Fe³⁺, Cr³⁺, Co²⁺, and Cu²⁺ metal ions. Interestingly, the order of stabilization energies, $E^{(2)}$, which indicates the greater charge transfer was found to be Ti⁴⁺ > Fe³⁺ > Cr³⁺ > Co²⁺ > Cu²⁺ which is consistent with the metal ion bioaffinities in Lf.

Conclusions

The observed trend for the metal ion binding energies in a vacuum and aqueous media in terms of metal affinity is Ti⁴⁺ > Fe³⁺ > Cr³⁺ > Co²⁺ > Cu²⁺ in reasonable agreement with the literature binding constant values ($\log k_{\text{binding}}$)⁵ and the reported ICP-MS results.²⁸

This work justifies the possible trend and electronic structural reasons for the competitive recognition of metal ion bioaffinities of Ti⁴⁺, Fe³⁺, Cr³⁺, and Co²⁺, and Cu²⁺ metal ions in human serum lactoferrin using density functional theory (DFT). The electronic properties such as frontier molecular obtained orbitals ($\Delta E_{\text{HOMO-LUMO}}$), chemical hardness (η) and softness (S) were found to be helpful to obtain a better insight into the molecular properties and stability of the metal binding pocket of lactoferrin which affects the extent of metal ion bioaffinities.

To the best of our knowledge, this work undertaken for the first time demonstrates that a combination of quantum chemical properties using DFT with ICP-MS and electronic absorption spectroscopy (EAS) for metal binding affinity offers a promising strategy for the sensing of different metal ions using metalloproteins. The outcome of this study provides

Table 5 The second-order perturbation energies, $E^{(2)}$ (kcal mol⁻¹), for the dominant donor → acceptor charge transfer between the Ti⁴⁺, Fe³⁺, Cr³⁺, Co²⁺, and Cu²⁺ metal ions and the ligating residues involved in the metal binding pocket of (Lf) obtained by B3LYP/6-31+G* and LANL2DZ basis sets for ions

Complexes	Donor	Acceptor	$E^{(2)}$ (kcal mol ⁻¹)
Lf-Ti ⁴⁺	σ_{C-O} (ASP60)	LP* (Ti ⁴⁺)	9.37
	σ_{C-O} (TYR92)	LP* (Ti ⁴⁺)	9.02
	σ_{C-O} (TYR192)	LP* (Ti ⁴⁺)	11.65
	σ_{C-N} (HIS253)	LP* (Ti ⁴⁺)	10.05
	σ_{C-O1} (CO ₃ ²⁻)	LP* (Ti ⁴⁺)	6.78
	σ_{C-O2} (CO ₃ ²⁻)	LP* (Ti ⁴⁺)	11.65
	LP (ASP60-O)	LP* (Ti ⁴⁺)	52.03
	LP (TYR92-O)	LP* (Ti ⁴⁺)	49.09
	LP (TYR192-O)	LP* (Ti ⁴⁺)	45.70
	LP (HIS253-N)	LP* (Ti ⁴⁺)	98.36
	LP (CO ₃ ²⁻)-O ₁	LP* (Ti ⁴⁺)	70.27
	LP (CO ₃ ²⁻)-O ₂	LP* (Ti ⁴⁺)	72.97
Lf-Fe ³⁺	σ_{C-O} (ASP60)	LP* (Fe ³⁺)	3.55
	σ_{C-O} (TYR92)	LP* (Fe ³⁺)	6.12
	σ_{C-O} (TYR192)	LP* (Fe ³⁺)	11.60
	σ_{C-N} (HIS253)	LP* (Fe ³⁺)	4.37
	σ_{C-O1} (CO ₃ ²⁻)	LP* (Fe ³⁺)	6.70
	σ_{C-O2} (CO ₃ ²⁻)	LP* (Fe ³⁺)	6.16
	LP (ASP60-O)	LP* (Fe ³⁺)	20.57
	LP (TYR92-O)	LP* (Fe ³⁺)	20.06
	LP (TYR192-O)	LP* (Fe ³⁺)	44.74
	LP (HIS253-N)	LP* (Fe ³⁺)	16.27
	LP (CO ₃ ²⁻)-O ₁	LP* (Fe ³⁺)	17.83
	LP (CO ₃ ²⁻)-O ₂	LP* (Fe ³⁺)	12.99
Lf-Cr ³⁺	σ_{C-O} (ASP60)	LP* (Cr ³⁺)	
	σ_{C-O} (TYR92)	LP* (Cr ³⁺)	3.92
	σ_{C-O} (TYR192)	LP* (Cr ³⁺)	4.49
	σ_{C-N} (HIS253)	LP* (Cr ³⁺)	6.69
	σ_{C-O1} (CO ₃ ²⁻)	LP* (Cr ³⁺)	5.77
	σ_{C-O2} (CO ₃ ²⁻)	LP* (Cr ³⁺)	18.15
	LP (ASP60-O)	LP* (Cr ³⁺)	15.55
	LP (TYR92-O)	LP* (Cr ³⁺)	44.64
	LP (TYR192-O)	LP* (Cr ³⁺)	12.16
	LP (HIS253-N)	LP* (Cr ³⁺)	17.31
	LP (CO ₃ ²⁻)-O ₁	LP* (Cr ³⁺)	12.03
	LP (CO ₃ ²⁻)-O ₂	LP* (Cr ³⁺)	
Lf-Co ²⁺	σ_{C-O} (ASP60)	LP* (Co ²⁺)	2.60
	σ_{C-O} (TYR92)	LP* (Co ²⁺)	6.10
	σ_{C-O} (TYR192)	LP* (Co ²⁺)	2.97
	σ_{C-N} (HIS253)	LP* (Co ²⁺)	2.87
	σ_{C-O1} (CO ₃ ²⁻)	LP* (Co ²⁺)	2.44
	σ_{C-O2} (CO ₃ ²⁻)	LP* (Co ²⁺)	5.52
	LP (ASP60-O)	LP* (Co ²⁺)	19.10
	LP (TYR92-O)	LP* (Co ²⁺)	15.50
	LP (TYR192-O)	LP* (Co ²⁺)	44.49
	LP (HIS253-N)	LP* (Co ²⁺)	16.10
	LP (CO ₃ ²⁻)-O ₁	LP* (Co ²⁺)	17.21
	LP (CO ₃ ²⁻)-O ₂	LP* (Co ²⁺)	3.33
Lf-Cu ²⁺	σ_{C-O} (ASP60)	LP* (Cu ²⁺)	2.51
	σ_{C-O} (TYR92)	LP* (Cu ²⁺)	5.52
	σ_{C-O} (TYR192)	LP* (Cu ²⁺)	0.61
	σ_{C-N} (HIS253)	LP* (Cu ²⁺)	2.27
	σ_{C-O1} (CO ₃ ²⁻)	LP* (Cu ²⁺)	4.36
	σ_{C-O2} (CO ₃ ²⁻)	LP* (Cu ²⁺)	5.40
	LP (ASP60-O)	LP* (Cu ²⁺)	15.52
	LP (TYR92-O)	LP* (Cu ²⁺)	15.07
	LP (TYR192-O)	LP* (Cu ²⁺)	44.06
	LP (HIS253-N)	LP* (Cu ²⁺)	15.44
	LP (CO ₃ ²⁻)-O ₁	LP* (Cu ²⁺)	13.42
	LP (CO ₃ ²⁻)-O ₂	LP* (Cu ²⁺)	2.02

helpful implications concerning the optimal metal ions, which can potentially act as binders in the metal binding pocket.

Further studies on the effect of the more distant coordination layer on the metal competition and comparison of the aforementioned concept between the lactoferrin and transferrin families are currently underway in our laboratory.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

BH gratefully acknowledges the financial support to undertake a postdoctoral position in the Computational Modelling and BioAnalytical Research Group in Chemistry at the Durban University of Technology (DUT). KB thanks the Centre for High Performance Computing (CHPC), Cape Town, South Africa, for all the computational resources.

References

- 1 G. Duca, *Homogeneous Catalysis with Metal Complexes: Fundamentals and Applications*, Berlin Heidelberg, 2012, vol. 478.
- 2 L. Balcaen, E. Bolea-Fernandez, M. Resano and F. Vanhaecke, Accurate determination of ultra-trace levels of Ti in blood serum using ICP-MS/MS, *Anal. Chim. Acta*, 2014, **809**, 1–8.
- 3 A. Sarmiento-Gonzalez, J. M. Marchante-Gayon, J. M. Tejerina-Lobo, J. Paz-Jimenez and A. Sanz-Medel, High-resolution ICP-MS determination of Ti, V, Cr, Co, Ni, and Mo in human blood and urine of patients implanted with a hip or knee prosthesis, *Anal. Bioanal. Chem.*, 2008, **391**, 2583–2589.
- 4 H. M. Baker, C. J. Baker, C. A. Smith and E. N. Baker, Metal substitution in transferrins: specific binding of cerium(IV) revealed by the crystal structure of cerium substituted human lactoferrin, *J. Biol. Inorg. Chem.*, 2000, **5**, 692–698.
- 5 H. Z. Sun, H. Y. Li and P. J. Sadler, Transferrin as a metal ion mediator, *Chem. Rev.*, 1999, **99**, 2817–2842.
- 6 L. Anghel, An investigation of the protonation states of human lactoferrin iron-binding protein, *Chem. J. Mold.*, 2015, **10**, 71–75.
- 7 P. Lisiecki, Transferrin and Lactoferrin – Human Iron Sources for Enterococci, *Pol. J. Microbiol.*, 2017, **66**, 419–425.
- 8 J. Dunham, J. Bauer, G. R. Campbell, D. J. Mahad, N. van Driel, S. M. A. van der Pol, B. A. t Hart, H. Lassmann, J. D. Laman, J. van Horssen and Y. S. Kap, Oxidative Injury and Iron Redistribution Are Pathological Hallmarks of Marmoset Experimental Autoimmune Encephalomyelitis, *J. Neuropathol. Exp. Neurol.*, 2017, **76**, 467–478.
- 9 A. V. Sokolov, K. V. Solovyov, V. A. Kostevich, A. V. Chekanov, M. O. Pulina, E. T. Zakharova, M. M. Shavlovski, O. M. Panasenkov and V. B. Vasilyev, Protection of ceruloplasmin by lactoferrin against hydroxyl radicals is pH dependent, *Biochem. Cell Biol.*, 2012, **90**, 397–404.

- 10 J. M. Steijns and A. C. M. van Hooijdonk, Occurrence, structure, biochemical properties and technological characteristics of lactoferrin, *Br. J. Nutr.*, 2000, **84**, S11–S17.
- 11 G. Deng, K. Wu, A. A. Cruce, M. K. Bowman and J. B. Vincent, Binding of trivalent chromium to serum transferrin is sufficiently rapid to be physiologically relevant, *J. Inorg. Biochem.*, 2015, **143**, 48–55.
- 12 L. Rosa, A. Cutone, M. S. Lepanto, M. J. Scotti, M. P. Conte, R. Paesano and P. Valenti, Physico-chemical properties influence the functions and efficacy of commercial bovine lactoferrins, *Biomaterials*, 2018, **31**, 301–312.
- 13 G. Majka, G. Wiecek, M. Srodek, K. Spiewak, M. Brindell, J. Koziel, J. Marcinkiewicz and M. Strus, The impact of lactoferrin with different levels of metal saturation on the intestinal epithelial barrier function and mucosal inflammation, *Biomaterials*, 2016, **29**, 1019–1033.
- 14 A. A. Moshtaghi, A. Badii, M. Fazilati, T. Hassanzadeh and H. Kharazi, Manganese and iron binding to human transferrin, *Iran. J. Pharm. Res.*, 2007, **6**, 101–106.
- 15 A. A. Moshtaghi and M. A. Ghaffari, Study of the Binding of Iron and Indium to Human Serum Apo-Transferrin, *Iran. Biomed. J.*, 2003, **7**, 73–77.
- 16 A. A. Moshtaghi, M. Ani and M. H. Arabi, Spectroscopic Studies on Titanium Ion Binding to the Apolactoferrin, *Iran. Biomed. J.*, 2006, **10**, 93–98.
- 17 H. J. Vogel, Lactoferrin, a bird's eye view introduction, *Biochem. Cell Biol.*, 2012, **90**, 233–244.
- 18 N. Saito, T. Iio, Y. Yoshikawa, H. Ohtsuka and K. Orino, Heme-binding of bovine lactoferrin: the potential presence of a heme-binding capacity in an ancestral transferrin gene, *Biomaterials*, 2018, **31**, 131–138.
- 19 K. J. Waldron and N. J. Robinson, How do bacterial cells ensure that metalloproteins get the correct metal?, *Nat. Rev. Microbiol.*, 2009, **7**, 25.
- 20 W. Neumann, R. C. Hadley and E. M. Nolan, Transition metals at the host–pathogen interface: how *Neisseria* exploit human metalloproteins for acquiring iron and zinc, in *Bioinorganic Chemistry*, ed. S. J. Lippard and J. M. Berg, 2017, pp. 211–223.
- 21 S. Sharma, M. Sinha, S. Kaushik, P. Kaur and T. P. Singh, C-Lobe of Lactoferrin: The Whole Story of the Half-Molecule, *Biochem. Res. Int.*, 2013, **2013**, 271641.
- 22 E. N. Baker and H. M. Baker, Molecular structure, binding properties and dynamics of lactoferrin, *Cell. Mol. Life Sci.*, 2005, **62**, 2531–2539.
- 23 L. Anghel, Lactoferrin: analysis of the structure profile, *Chem. J. Mold.*, 2014, **9**(2), 99–106.
- 24 L. A. Lambert, H. Perri, P. J. Halbrooks and A. B. Mason, Evolution of the transferrin family: conservation of residues associated with iron and anion binding, *Comp. Biochem. Physiol., Part B: Biochem. Mol. Biol.*, 2005, **142**, 129–141.
- 25 B. F. Andersen, H. M. Baker, G. E. Morris, S. V. Rumball and E. N. Baker, Apolactoferrin structure demonstrates ligand-induced conformational change in transferrins, *Nature*, 1990, **344**, 784–787.
- 26 M. P. Chantada-Vazquez, A. Moreda-Pineiro, M. C. Barciela-Alonso and P. Bermejo-Barrera, Spectrometric-based techniques for metal-binding protein assessment in clinical, environmental, and food samples, *Appl. Spectrosc. Rev.*, 2017, **52**, 145–174.
- 27 C. Izzo, Z. A. Doubleday and B. M. Gillanders, Where do elements bind within the otoliths of fish?, *Mar. Freshwater Res.*, 2016, **67**, 1072–1076.
- 28 K. Loeschner, C. F. Harrington, J. L. Kearney, D. J. Langton and E. H. Larsen, Feasibility of asymmetric flow field-flow fractionation coupled to ICP-MS for the characterization of wear metal particles and metalloproteins in biofluids from hip replacement patients, *Anal. Bioanal. Chem.*, 2015, **407**, 4541–4554.
- 29 A. Motahari and A. Fattahi, Theoretical aspects of the enhancement of metal binding affinity by intramolecular hydrogen bonding and modulating $pK_{(a)}$ values, *New J. Chem.*, 2017, **41**, 15110–15119.
- 30 C. Sill, R. Biehl, B. Hoffmann, A. Radulescu, M. S. Appavou, B. Farago, R. Merkel and D. Richter, Structure and domain dynamics of human lactoferrin in solution and the influence of Fe(III)-ion ligand binding, *BMC Biophys.*, 2016, **9**, 7.
- 31 M. Acosta, S. Torres, L. Marino-Repizo, L. D. Martinez and R. A. Gil, Novel method for metalloproteins determination in human breast milk by size exclusion chromatography coupled to inductively coupled plasma mass spectrometry, *J. Pharm. Biomed. Anal.*, 2018, **158**, 209–213.
- 32 J. S. Garcia, C. S. de Magalhaes and M. A. Z. Arruda, Trends in metal-binding and metalloprotein analysis, *Talanta*, 2006, **69**, 1–15.
- 33 J. C. Swarts, P. J. Swarts, E. N. Baker and H. M. Baker, Spectroscopic studies on metal complexes of *Aspergillus awamori*-derived recombinant human lactoferrin, *Inorg. Chim. Acta*, 2000, **298**, 187–194.
- 34 F. Superti, M. Agamennone, A. Pietrantonio and M. G. Ammendolia, Bovine Lactoferrin Prevents Influenza A Virus Infection by Interfering with the Fusogenic Function of Viral Hemagglutinin, *Viruses*, 2019, **11**, 51.
- 35 S. E. Soboleva, T. A. Guschina and G. A. Nevinsky, Human serum and milk albumins are metal-dependent DNases, *IUBMB Life*, 2018, **70**, 501–510.
- 36 X. K. Cai, Y. F. Duan, Y. Li, J. Wang, Y. Y. Mao, Z. Y. Yang, X. F. Zhao, Y. R. Zhao, Y. Guan and S. Yin, Lactoferrin level in breast milk: a study of 248 samples from eight regions in China, *Food Funct.*, 2018, **9**, 4216–4222.
- 37 A. V. Sokolov, I. V. Voynova, V. A. Kostevich, A. Y. Vlasenko, E. T. Zakharova and V. B. Vasilyev, Comparison of interaction between ceruloplasmin and lactoferrin/transferrin: to bind or not to bind, *Biochemistry*, 2017, **82**, 1073–1078.
- 38 J. Y. Huang, T. Liu, J. Cao, Z. J. Wu, X. Y. Liao and H. S. Song, Electrochemical Studies on Bactericidal Mechanism of Human Lactoferrin and Its Demand for Microenvironment, *Int. J. Electrochem. Sci.*, 2017, **12**, 1445–1452.
- 39 F. Carmona, V. Munoz-Robles, R. Cuesta, N. Galvez, M. Capdevila, J. D. Marechal and J. M. Dominguez-Vera, Monitoring lactoferrin iron levels by fluorescence resonance energy transfer: a combined chemical and computational study, *J. Biol. Inorg. Chem.*, 2014, **19**, 439–447.
- 40 I. A. Garcia-Montoya, T. S. Cendon, S. Arevalo-Gallegos and Q. Rascon-Cruz, Lactoferrin a multiple bioactive protein: an

- overview, *Biochim. Biophys. Acta, Gen. Subj.*, 2012, **1820**, 226–236.
- 41 X. L. Sun, H. M. Baker, S. C. Shewry, G. B. Jameson and E. N. Baker, Structure of recombinant human lactoferrin expressed in *Aspergillus awamori*, *Acta Crystallogr., Sect. D: Biol. Crystallogr.*, 1999, **55**, 403–407.
 - 42 M. Lanznaster, A. Neves, A. J. Bortoluzzi, A. M. C. Assumpcao, I. Vencato, S. P. Machado and S. M. Drechsel, Electronic effects of electron-donating and -withdrawing groups in model complexes for iron-tyrosine-containing metalloenzymes, *Inorg. Chem.*, 2006, **45**, 1005–1011.
 - 43 B. Lonnerdal, Trace element nutrition of infants – molecular approaches, *J. Trace Elem. Med. Biol.*, 2005, **19**, 3–6.
 - 44 P. N. Sunilkumar, C. Sadasivan, K. S. Devaky and M. Haridas, Conformation of an octapeptide fragment (2–9) of kaliocin-1 in DMSO-d(6) by H-1 NMR and restrained molecular dynamics, *Ind. J. Biochem. Biophys.*, 2007, **44**, 44–49.
 - 45 M. Umeyama, A. Kira, K. Nishimura and A. Naito, Interactions of bovine lactoferricin with acidic phospholipid bilayers and its antimicrobial activity as studied by solid-state NMR, *Biochim. Biophys. Acta, Biomembr.*, 2006, **1758**, 1523–1528.
 - 46 W. G. Jing, J. S. Svendsen and H. J. Vogel, Comparison of NMR structures and model-membrane interactions of 15-residue antimicrobial peptides derived from bovine lactoferricin, *Biochem. Cell Biol.*, 2006, **84**, 312–326.
 - 47 R. T. A. MacGillivray, S. A. Moore, J. Chen, B. F. Anderson, H. Baker, Y. Luo, M. Bewley, C. A. Smith, M. E. P. Murphy, Y. Wang, A. B. Mason, R. C. Woodworth, G. D. Brayer and E. N. Baker, Two High-Resolution Crystal Structures of the Recombinant N-Lobe of Human Transferrin Reveal a Structural Change Implicated in Iron Release, *Biochemistry*, 1998, **37**, 7919–7928.
 - 48 H. R. Faber, T. Bland, C. Day, G. Norris, J. W. Tweedie and E. Baker, Altered Domain Closure and Iron Binding in Transferrins: The Crystal Structure of the Asp60Ser Mutant of the Amino-terminal Half-molecule of Human Lactoferrin, *J. Mol. Biol.*, 1996, **256**, 352–363.
 - 49 J. I. Mujika, X. Lopez, E. Rezabal, R. Castillo, S. Marti, V. Moliner and J. M. Ugalde, A QM/MM study of the complexes formed by aluminum and iron with serum transferrin at neutral and acidic pH, *J. Inorg. Biochem.*, 2011, **105**, 1446–1456.
 - 50 C. A. Smith, B. F. Anderson, H. M. Baker and E. N. Baker, Metal substitution in transferrins: the crystal structure of human copper-lactoferrin at 2.1 Å resolution, *Biochemistry*, 1992, **31**, 4527–4533.
 - 51 J. Hirose, H. Fujiwara, T. Magarifuchi, Y. Iguti, H. Iwamoto, S. Kominami and K. Hiromi, Copper binding selectivity of N- and C-sites in serum (human)- and ovo-transferrin, *Biochim. Biophys. Acta, Protein Struct. Mol. Enzymol.*, 1996, **1296**, 103–111.
 - 52 T. Sakajiri, H. Yajima and T. Yamamura, Density Functional Theory Study on Metal-Binding Energies for Human Serum Transferrin-Metal Complexes, *ISRN Biophys.*, 2012, **2012**, 5.
 - 53 C. Smith, B. Anderson, H. M. Baker and E. Baker, Structure of copper- and oxalate-substituted human lactoferrin at 2.0 Å resolution, *Acta Crystallogr., Sect. D: Biol. Crystallogr.*, 1994, **50**, 302–316.
 - 54 H. M. Berman, J. Westbrook, Z. Feng, G. Gilliland, T. N. Bhat, H. Weissig, I. N. Shindyalov and P. E. Bourne, The protein data bank, 1999–, in *International Tables for Crystallography Volume F: Crystallography of biological macromolecules*, 2006, Springer, pp. 675–684.
 - 55 K. Loeschner, C. F. Harrington, J. L. Kearney, D. J. Langton and E. H. Larsen, Feasibility of asymmetric flow field-flow fractionation coupled to ICP-MS for the characterization of wear metal particles and metalloproteins in biofluids from hip replacement patients, *Anal. Bioanal. Chem.*, 2015, **407**, 4541–4554.
 - 56 J. C. Gordon, J. B. Myers, T. Folta, V. Shoja, L. S. Heath and A. Onufriev, H⁺⁺: a server for estimating pK_(a)s and adding missing hydrogens to macromolecules, *Nucleic Acids Res.*, 2005, **33**, W368–W371.
 - 57 R. G. Parr and W. Yang, *Density Functional Theory of Atoms and Molecules*, Oxford Univ Press, Oxford, 1989.
 - 58 M. J. Frisch, G. W. Trucks, H. B. Schlegel, G. E. Scuseria, M. A. Robb, J. R. Cheeseman, G. Scalmani, V. Barone, G. A. Petersson, H. Nakatsuji, X. Li, M. Caricato, A. V. Marenich, J. Bloino, B. G. Janesko, R. Gomperts, B. Mennucci, H. P. Hratchian, J. V. Ortiz, A. F. Izmaylov, J. L. Sonnenberg, D. Y. Williams, F. Ding, F. Lipparini, F. Egidi, J. Goings, B. Peng, A. Petrone, T. Henderson, D. Ranasinghe, V. G. Zakrzewski, J. Gao, N. Rega, G. Zheng, W. Liang, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, T. Vreven, K. Throssell, J. A. Montgomery Jr., J. E. Peralta, F. Ogliaro, M. J. Bearpark, J. J. Heyd, E. N. Brothers, K. N. Kudin, V. N. Staroverov, T. A. Keith, R. Kobayashi, J. Normand, K. Raghavachari, A. P. Rendell, J. C. Burant, S. S. Iyengar, J. Tomasi, M. Cossi, J. M. Millam, M. Klene, C. Adamo, R. Cammi, J. W. Ochterski, R. L. Martin, K. Morokuma, O. Farkas, J. B. Foresman and D. J. Fox, *Gaussian 16 Rev. B.01*, Wallingford, CT, 2016.
 - 59 T. H. Dunning Jr. and P. J. Hay, in *Methods of Electronic Structure Theory*, ed. H. F. Schaefer III, Plenum Press, 1977, vol. 2, pp. 1–28.
 - 60 P. J. Hay and R. Wadt Willard, *Ab initio* effective core potentials for molecular calculations. Potentials for K to Au including the outermost core orbitals, *J. Chem. Phys.*, 1985, **82**, 299–310.
 - 61 V. Coropceanu, M. Malagoli, D. da Silva Filho, N. Gruhn, T. Bill and J. Brédas, Hole- and electron-vibrational couplings in oligoacene crystals: intramolecular contributions, *Phys. Rev. Lett.*, 2002, **89**, 275503.
 - 62 F. Y. Adeowo, B. Honarparvar and A. A. Skelton, The interaction of NOTA as a bifunctional chelator with competitive alkali metal ions: a DFT study, *RSC Adv.*, 2016, **6**, 79485–79496.
 - 63 F. Y. Adeowo, B. Honarparvar and A. A. Skelton, Density Functional Theory Study on the Complexation of NOTA as a Bifunctional Chelator with Radiometal Ions, *J. Phys. Chem. A*, 2017, **121**, 6054–6062.

- 64 E. Frimpong, A. A. Skelton and B. Honarparvar, DFT study of the interaction between DOTA chelator and competitive alkali metal ions, *J. Mol. Graph. Modell.*, 2017, **76**, 70–76.
- 65 A. Skelton, N. Agrawal and J. Fried, Quantum Mechanical Calculations of the Interactions Between Diazacrowns and the Sodium Cation: an insight into Na⁺ Complexation in Diazacrown-Based Synthetic Ion Channels, *RSC Adv.*, 2015, **5**, 55033–55047.
- 66 S. Miertus and J. Tomasi, Approximate evaluations of the electrostatic free energy and internal energy changes in solution processes, *Chem. Phys.*, 1982, **65**, 239–245.
- 67 J. Tomasi, B. Mennucci and E. Cancès, The IEF version of the PCM solvation method: an overview of a new method addressed to study molecular solutes at the QM *ab initio* level, *THEOCHEM*, 1999, **464**, 211–226.
- 68 G. Scalmani and M. J. Frisch, Continuous surface charge polarizable continuum models of solvation. I. General formalism, *J. Chem. Phys.*, 2010, **132**, 114110.
- 69 D. T. Bowron, M. Amboage, R. Boada, A. Freeman, S. Hayama and S. Díaz-Moreno, The hydration structure of Cu²⁺: more tetrahedral than octahedral?, *RSC Adv.*, 2013, **3**, 17803–17812.
- 70 F. Weinhold and J. E. Carpenter, *The Structure of Small Molecules and Ions*, Plenum, New York, 1988.
- 71 J. P. Foster and F. Weinhold, Natural hybrid orbitals, *J. Am. Chem. Soc.*, 1980, **102**, 7211–7218.
- 72 A. E. Reed and F. Weinhold, Natural bond orbital analysis of analysis of near-Hartree–Fock water dimer, *J. Chem. Phys.*, 1983, **78**, 4066–4073.
- 73 T. Koopmans, Über die Zuordnung von Wellenfunktionen und Eigenwerten zu den Einzelnen Elektronen Eines Atoms, *Phys. Chem. Chem. Phys.*, 1933, **1**, 104–113.
- 74 R. G. Parr and R. G. Pearson, Absolute hardness: companion parameter to absolute electronegativity, *J. Am. Chem. Soc.*, 1983, **105**, 7512–7516.
- 75 G. F. Li, J. J. Wang, X. M. Chen, H. W. Yang, B. Yang, B. Q. Xu and D. C. Liu, Density functional investigation on structural and electronic properties of small bimetallic Pb-*n* Ag-*n* (*n* = 2–12) clusters, *J. Cent. South Univ.*, 2018, **25**, 772–782.
- 76 R. G. Pearson, Chemical hardness and density functional theory, *J. Chem. Sci.*, 2005, **117**, 369–377.
- 77 P. W. Ayers, R. G. Parr and R. G. Pearson, Elucidating the hard/soft acid/base principle: a perspective based on half-reactions, *J. Chem. Phys.*, 2006, **124**, 194107.
- 78 R. G. Pearson, The electronic chemical potential and chemical hardness, *THEOCHEM*, 1992, **255**, 261–270.
- 79 W. Kohn and L. J. Sham, Self-Consistent Equations Including Exchange and Correlation Effects, *Phys. Rev.*, 1965, **140**, A1133–A1138.
- 80 S. G. Patra, Lewis acidity of benzene in half-sandwich ruthenium arene complex. A computational study, *Comput. Theor. Chem.*, 2018, **1136**, 34–48.
- 81 J. Lameira, C. N. Alves, V. Moliner and E. Silla, A density functional study of flavonoid compounds with anti-HIV activity, *Eur. J. Med. Chem.*, 2006, **41**, 616–623.
- 82 J. Lameira, I. G. Medeiros, M. Reis, A. S. Santos and C. N. Alves, Structure–activity relationship study of flavone compounds with anti-HIV-1 integrase activity: a density functional theory study, *Bioorg. Med. Chem.*, 2006, **14**, 7105–7112.
- 83 A. E. Reed, L. A. Curtiss and F. Weinhold, Intermolecular interactions from a natural bond orbital, donor–acceptor viewpoint, *Chem. Rev.*, 1988, **88**, 899–936.
- 84 A. E. Reed, L. A. Curtiss and F. Weinhold, Intermolecular interactions from a natural bond orbital, donor–acceptor viewpoint, *Chem. Rev.*, 1988, **88**, 899–926.
- 85 W. R. Harris, A. M. Cafferty, S. Abdollahi and K. Trankler, Binding of monovalent anions to human serum transferrin, *Biochim. Biophys. Acta, Protein Struct. Mol. Enzymol.*, 1998, **1383**, 197–210.
- 86 S. A. Abdel-Latif and H. Moustafa, Synthesis, spectroscopic properties, density functional theory calculations and non-linear optical properties of novel complexes of 5-hydroxy-4,7-dimethyl-6-(phenylazo)coumarin with Mn(II), Co(II), Ni(II), Cu(II) and Zn(II) metal ions, *Appl. Organomet. Chem.*, 2018, **32**.
- 87 J.-n. Liu, Z.-r. Chen and S.-f. Yuan, Study on the prediction of visible absorption maxima of azobenzene compounds, *J. Zhejiang Univ., Sci., B*, 2005, **6**, 584.
- 88 S. Sebastian, S. Sylvestre, J. Jayabharathi, S. Ayyapan, M. Amalanathan, K. Oudayakumar and I. A. Herman, Study on conformational stability, molecular structure, vibrational spectra, NBO, TD-DFT, HOMO and LUMO analysis of 3,5-dinitrosalicylic acid by DFT techniques, *Spectrochim. Acta, Part A*, 2015, **136**(Part B), 1107–1118.
- 89 M. Szafran, A. Komasa and E. Bartoszak-Adamska, Crystal and molecular structure of 4-carboxypiperidinium chloride (4-piperidinecarboxylic acid hydrochloride), *J. Mol. Struct.*, 2007, **827**, 101–107.
- 90 E. Kavitha, N. Sundaraganesan, S. Sebastian and M. Kurt, Molecular structure, anharmonic vibrational frequencies and NBO analysis of naphthalene acetic acid by density functional theory calculations, *Spectrochim. Acta, Part A*, 2010, **77**, 612–619.
- 91 T. Rajamani and S. Muthu, Electronic absorption, vibrational spectra, non-linear optical properties, NBO analysis and thermodynamic properties of 9-[(2-hydroxyethoxy)methyl] guanine molecule by density functional method, *Solid State Sci.*, 2013, **16**, 90–101.